Optimization Approach for a SAR small satellite in Very Low Earth Orbit (VLEO) Demonstrator Mission

Yago Molano Gomez
ST Engineering Satellite Systems
Imperial College London
London, United Kingdom
yagomolanogomez@gmail.com

Abstract— **This study provides an in-depth exploration into the design, optimization, and feasibility of deploying Synthetic Aperture Radar (SAR) satellites in Very Low Earth Orbit (VLEO). Leveraging the inherent advantages of VLEO, such as increased payload capacity and higher resolution imaging, the study aims to overcome the typical challenges of atmospheric drag and system stability through innovative design and optimization techniques. A comprehensive optimization tool was developed to iteratively refine the satellite's shape, subsystems, and component choices, taking into account aerodynamic considerations unique to VLEO. Remarkably, the optimized design not only meets but exceeds the set mission objectives, achieving a total system mass well below the 150kg target and an imaging resolution as fine as 0.25m. While challenges related to aerodynamic effects and system stability exist, they are far from insurmountable and expected to diminish with technological advancements. The optimization tool developed is versatile enough that could be adapted for different payloads and orbits, offering a generalizable approach that could accelerate advancements in earth observation.**

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1. INTRODUCTION

Advantages and Challenges of VLEO

Very Low Earth Orbit (VLEO) offers unique advantages largely due to its proximity to Earth. One of the most significant benefits is the enhanced resolution capabilities of optical payloads. This closer orbit allows for either an increase in imaging resolution or the use of smaller, lighter optical payloads to achieve the same level of resolution as higher orbits. In terms of surveillance, VLEO provides a wider maximum beamwidth due to a reduced travel path for sensor signals, which is particularly advantageous for wide-area monitoring. Launching into VLEO also presents economic advantages. By reducing the target orbit altitude from the typical 600km to 300km, launchers can carry more payload mass—up to 13% more for the VEGA launcher and as much as double for the Dnepr launch vehicle [1]. This increased payload capacity makes VLEO an attractive option for piggyback or secondary missions.

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However, VLEO is not without its challenges, most of which stem from the denser atmospheric conditions. Satellites in VLEO experience higher drag forces, increasing the likelihood of orbital decay within a five-year period unless compensated for. The presence of atomic oxygen at this altitude poses additional risks, including oxidation and corrosion of sensitive materials. Specialized aerodynamic designs and materials that can resist these conditions are therefore essential [2].

Interestingly, the very challenges of VLEO also offer some advantages. The increased drag eliminates the need for a deorbit maneuver, saving propellant. Furthermore, the lower altitude has less space debris and a rapid decay rate for any objects, significantly lowering the risk of collision. However, it should be noted that the reduced altitude also shortens the communication window, requiring higher bandwidth and potentially increasing the mass and power requirements of the communication subsystem.

SAR in VLEO

For Synthetic Aperture Radar (SAR) payloads, VLEO offers a range of benefits similar to those for optical payloads. The closer proximity to Earth enables higher resolution imaging without requiring hardware modifications. Additionally, the power density of the signal improves with the inverse square of the distance, leading to a higher signal-to-noise ratio. These advantages contribute to a virtuous cycle where reductions in power and sensor area subsequently reduce the size and power requirements for other subsystems, including solar panels.

Mission objectives and requirements

The satellite was designed in alignment with the following mission statement: "Demonstrator mission of a small satellite in very low Earth Orbit (VLEO) to investigate its feasibility, benefits and drawbacks when compared to traditional LEO satellites, for an earth observation SAR payload".

The mission objectives were to:

- Identify suitable shapes and sizes for the satellite bus, to minimize power consumption during operation.
- Identify power requirements.
- Compare power requirements, materials and cost to a satellite in LEO.
- Evaluate the feasibility and technical requirements for a SAR payload in the VLEO environment.
- Maximize payload fraction mass, especially when compared with a standard 150kg in LEO.

The requirements for the mission are displayed in Table 1

Initially, project requirements suggested a near-equatorial

Table 1. Requirements for the given mission

Requirement	Value
Mass	$\leq 150kg$
Inclination	$10°$
Altitude	$250km$
Eccentricity	Sun-synchronous
Mission design life	5 years
EO Performance	$< 1m$ resolution
Attitude	$< 0.1°$ pointing accuracy
Survivability	Survive atomic oxygen bombardment of environment at 250km altitude for duration of the mission. Relaxed radiation reqs due to alt.

orbit. However, a trade study conducted during the project's inception revealed that a sun-synchronous orbit offered a more efficient design point, primarily due to the required sizing of batteries and solar panels. In a sun-synchronous orbit, continuous sun exposure ensures only minor variations in energy absorption by the solar panels, depending on their alignment with the sun's direction. In contrast, a near-equatorial orbit would lead to solar panels absorbing approximately double the energy required, necessitating storage by batteries during each orbit. Opting for a sun-synchronous orbit significantly reduces both the battery and solar panel weights. Importantly, this choice has no adverse impact on the Synthetic Aperture Radar (SAR) payload since it operates independently of natural light sources, making it immune to the constant dusk conditions in such an orbit.

Regarding mission lifespan, a 5-year duration was selected. This choice was made considering the substantial mass of fuel required to sustain a very low Earth orbit (VLEO). Extending the mission's design life significantly would impose severe mass penalties. The 5-year timeframe was deemed ambitious yet feasible, especially when compared to existing VLEO missions, whilst being long enough to attract commercial interest.

Mission architecture

The key elements of the selected mission architecture are shown in Table 2. The table focuses on those that could differ from a conventional LEO small satellite.

For the propulsion, both cold gas and chemical propulsion were discarded as they were unfeasible for the delta-v's expected. Between electric propulsion (EP) and air-breathing electric propulsion (ABEP), ABEP is considered the best option, but EP was chosen due to technology readiness.

Regarding the power source, solar panels were considered optimal. Body mounted was chosen to minimize the cross-sectional area of the satellite and to achieve a symmetrical, aerodynamically stable design.

Regarding geometry, an initial wedge-like design was chosen, as from the literature described in section 3 it should provide minimal drag. This design was iterated upon and optimized.

Regarding ADCS, it was assumed that reaction wheels and a magnetorquer could provide adequate control, especially considering that the design can be made aerodynamically stable by making sure the centre of mass is ahead of the centre of pressure. In case this was not enough, aerodynamic actuators could be considered.

2. SAR TRADE STUDY

At the initial stage, the SAR was defined assuming its parameters, so that a comparison could be carried out between VLEO and LEO.

Assumptions and target requirements set

As most values vary per platform, for the initial assumption stage the value of an existing platform was assumed [2]. These are shown in Table 3.

The target along-track resolution, across-track resolution, range and target Noise Equivalent Sigma Zero (NESZ) were defined and are shown in Table 4. In a SAR payload, the along-track resolution constraints the length of the panel for non-ambiguous results. Because of this, it tends to be coarser than desired, but will be improved when in squint and spotlight modes.

Results

The SAR code was run for a number of areas and dimensions, and altitudes. This initial model was based on the SAR Power Formula as described in literature [1], and did not take into account the sensor's architecture. This rough approximation was validated against the values in the referenced paper [1], and by comparing them with peak power values for the payloads of previous missions by ST Engineering.

Figure 1 shows the relationship between area and power to achieve the target parameters. The bars come exponentially closer for small areas, and the theoretical limit is reached a short distance below the 1400W line. A SAR platform with an area of $2m^2$ would need around 400W of power to run at an altitude of 250km. For an altitude of 550km, the power would be prohibitively expensive and maybe even impossible due to the minimum area for non-ambiguous results. An area of about $3m^2$ would be required with a power of 1400W, already leading to a 50% increase in SAR mass before accounting for the effects of the extra power.

This phenomenon is further illustrated in Figure 2. A resolution of $1m$, which is currently considered very good, can be achieved with a peak power of $200W$ for an area of $1.5m^2$. At $500km$, a resolution half as good (of $2m$) has a peak power of $500W$, showing how exponential the altitude penalty is for this type of payload.

Table 2. Satellite configuration selected for the mission.

Mission element	Possible options for given mission
Sensor	Phased array
Size	Small aperture
Propulsion	Electric propulsion
Attitude	3-axis stabilization: gyros, star sensor, sun sensor
Power source	Body mounted solar
Geometry	Wedge
Control actuators	Magnetic torque and reaction wheels

Table 3. Assumed parameters for the SAR antenna. [2]

Assumed parameter	Value
Antenna area density	10 kg/m^2
Signal wavelength	31.1mm (X-band)
Antenna efficiency	0.5
System loss	2.6dB
Noise figure	4 dB
Receiver noise temperature	700K

Table 4. SAR Target Requirements

Target requirement	Value
Along-track resolution	1.5m
Across-track resolution	0.25m
Range	25km^2
Target NESZ	-20dB

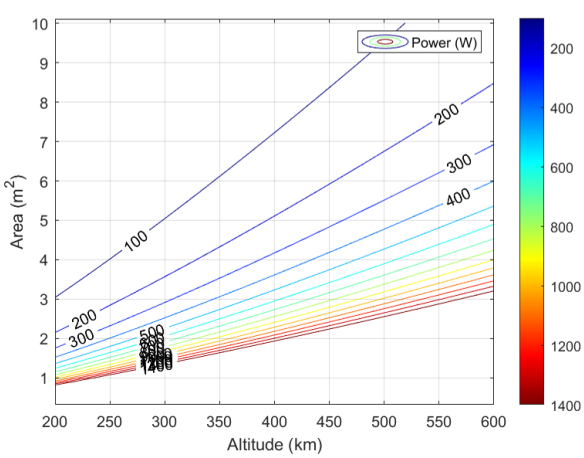

Figure 1. Antenna area required at each altitude to achieve the target parameters in Table 4, as well as the power required to achieve it.

These two examples clearly illustrate the advantages a VLEO can provide for the SAR payload.

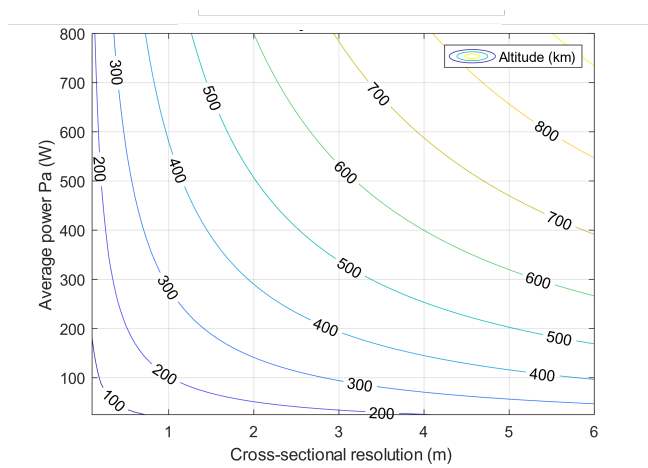

Figure 2. Relation between power and resolution for all altitudes, for an arbitrary SAR antenna area of 1.5m^2

Improvements to SAR code

Whilst the initial SAR model included the power formula, it did not take into account other important characteristics which could affect the power, nor did it take into account how the noise losses vary in each situation.

Because of this, the SAR model was improved to characterize the SAR subsystem completely, using MATLAB's Radar Toolbox and methods in the literature [3, 4]. The Radar Toolbox in MATLAB is a comprehensive set of functions and tools designed for radar system design, simulation, and signal processing. It enables users to model and analyze radar systems, simulate radar signals and targets, and implement algorithms for tasks such as target detection, tracking, and range estimation. It also includes documentation and examples on how to apply it to a range of problems [5].

Firstly, the swath width was used to find the required grazing angle for the given height using a curved Earth model. To achieve the required across-track ground resolution, the required slant range resolution was found, which defined the required bandwidth. The slant range, swath width and azimuth resolution were used to find the PRF Bounds. The required azimuth resolution and synthetic aperture length defined the integration time. All of these values were then used in the power formula, as well as noise losses found using MATLAB's radar toolbox and including surface roughness

scattering, rain reflectivity, processing gains and atmospheric losses.

3. AERODYNAMIC ANALYSIS

Geometric shape to reduce drag

The largest factor in drag reduction is to minimise the cross-sectional area of a body, which can be explained by the simple drag equation. Once cross-sectional area has been minimised, further methods may be explored. Based on literature [6], drag can be reduced with two different methods, depending on the total length of the body compared to its height. For a spacecraft carrying a SAR sensor, it is most likely that its shape will be elongated, and therefore the method for longer rectangles should be considered. For an ideal reduction in C_D coefficient, the best option is given by a cone which is larger than half the height of the design with no nose. If this is constrained, then the optimal is harder to gauge as it depends on the angle of the cone, as well as the size of its nose, as shown by Figure 3.

Fixed Cone Length (m)	Optimum Body for Minimum Drag	C_D (% reduction on no cone)	Volume [m^3]
No Nose Cone		2.72	1.00
0.1		2.62 (3.5%)	0.99 (1%)
0.2		2.55 (6.3%)	0.96 (4%)
0.3		2.47 (9.1%)	0.93 (7%)

Figure 3. **Effect of nose on drag reduction [6]**

Initial shape chosen

The initial shape was chosen considering the need for an aerodynamic shape and positioning of the main components. Due to the sun-synchronous orbit, the solar panels are required to be placed at perpendicular to the downwards direction. The earth-observation payload must be placed pointing nearly downwards, with a small angle from the downwards direction. Meanwhile, the cross-sectional area must be minimal, and a symmetrical shape must be used so that the shape is aerodynamically stable. Because of this, a triangular cross-section was chosen for the initial design point, to have a small angle for the solar panels, the desired angle for the payload, and minimising cross-sectional area. A nose was initially included to reduce drag. The mesh of the satellite can be found in Figure 4.

Aerodynamic analysis using panel code method

During the initial sizing, there were concerns that the usually assumed C_D value of 2.2 for a boxy satellite could not apply in VLEO. This is because in VLEO the residual atmosphere can be considered highly rarified and therefore free molecular flow [2], which could result in the C_D value being an underestimate.

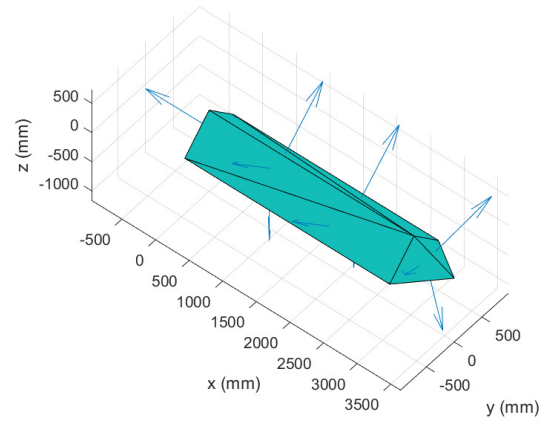

Figure 4. **Mesh showing initial concept for design.**

Table 5. **Initial C_D values obtained from ADBsat for the parameters in table 6.**

Geometry	C_D Value	C_m Magnitude
No nose	4.016	0.1189
Pointed nose ($l = 0.5h$)	3.965	0.1088

In order to test this, it was attempted to apply the 'ADBsat' MATLAB toolkit developed by the University of Manchester, with the latest design point to better understand the C_D values that could be expected. This toolkit allows for the calculation of C_D coefficient by inputting a .step file and the material properties, and computes them for free-molecular flow conditions with different gas-surface interaction models [7]. The selected equations to use with ADBsat's panel method were Sentman's, which assumes diffuse re-emission, and which uses a more realistic model for the velocity distribution of the incoming particles when compared to more simplistic models, and is therefore widely used for satellite flight in VLEO [8]. The model was employed assuming complete diffusion, and an energy accommodation factor of 1. It is expected that depending on the material used, this value could decrease, which would decrease the drag coefficient. This is therefore the upper bound value, a good conservative estimate for the initial stage.

A CAD of the mass optimization in Table 6 was made. Then, the ADBsat code was adapted to be compatible with the given design. The initial results can be found in Table 5. Afterwards, a script that could create the MATLAB mesh as a function of the shape parameters was made, so that the aerodynamic analysis could be included in the optimization, as seen in Figure 7.

4. OPTIMIZATION APPROACH

Sun-synchronous orbit problem

As previously described, this is no longer a straightforward problem where one variable should be maximised and the

others minimised, as all the subsystems are interlinked and changing the shape is advantageous for some subsystems but challenging for others. This can be understood intuitively by looking at the cross-section of the satellite shown in Figure 5. If the problem is only considered in 2D, there are two variables: height, which if reduced, decreases cross-sectional area, but increases required solar panel area by increasing sun angle of incidence; and width, which if increased reduces the power of the SAR, but also reduces the angle of incidence of the sun increasing the required solar panels, whilst making the cross-sectional area smaller.

If all of the remaining subsystems are included, it becomes even less intuitive; an increase in power could increase the size of the batteries and the thermal management system and also lead to an increase in the required solar panel size. Apart from the significant increase in mass, as the solar panels are body mounted for stability considerations, this could lead to an increase in cross-sectional area, leading to an increase in fuel and propulsive unit mass, and a further increase in power. Due to the vicious cycle, the gradient away from the optimal design point is steep, and it is therefore imperative that it was found so that the mass requirements were satisfied.

Initial design point

An initial shape optimization was carried out only considering the power and mass of the SAR, propulsion unit, solar panels and battery, and the cross-sectional area for drag calculations. This is shown in diagram form in Figure 6.

Efficiently managing mass in the optimization process poses a significant challenge, primarily due to the need for an initial estimate of structural mass. To establish this estimate, an empirical approach was employed, combining the recommended value from SMAD [9], with data from satellites of similar sizes. This empirical analysis yielded an initial estimate of structural mass at 62.5 kg. Subsequently, this value was divided by the surface area of the initial design configuration, enabling its scalability with respect to dimensions. This resulted in a 'surface density' metric of 8.39 kg/m², which serves as the basis for calculating structural mass during the optimization process.

It is important to note that while this surface density estimate may lack accuracy, it provides a reasonable starting point for the optimization. Importantly, it guides the optimization algorithm to recognize the correlation between increased dimensions and higher structural mass, thereby facilitating the optimization process. The results can be seen in Table 6. The initial mass optimization only includes the estimated mass of the solar panels, SAR and the fuel weight.

Optimizing for mass results in a smaller SAR area, causing a significant increase in power requirements. This prompts a height increase, but solar panel area expansion remains minimal. The cross-sectional area rises, leading to increased drag, with propulsion power only rising by 3W. In the end, it results in a mass reduction. However, this optimization doesn't account for potential complexities, such as the added weight from batteries and the thermal management system due to increased power, or the constraints imposed by higher drag on the choice of propulsion unit. Consequently, after demonstrating the optimization's superiority over a traditional iterative approach, all subsystems were integrated, and the algorithm was fine-tuned.

Current optimization algorithm

After a simpler optimization algorithm was tested, more subsystems, with more nuanced code, were added to the algorithm. The current optimization algorithm can be understood from the diagram in Figure 7.

From the resolution parameters, SARsizing calculates its gains, optimum PRF, and losses as well as the minimum area and minimum length required with the SAR to generate non-ambiguous images. The area is then used as a constraint, whilst the length is fed into the aerodynamic calculations, and is used to make a mesh optimization expression, only in terms of the variables of width, height, and noselength.

After the mesh is done, it creates an optimization expression for the C_D, which is fed into the delta-v budget block, whilst the expressions for aerodynamic moments and solar radiation pressure are fed into the ADCS subsystem. The delta-v budget creates a drag expression and a delta-v required expression and feeds that into the ADCS and the propulsion subsystem respectively. Aerodynamic calculations also give the surface area for each panel in the mesh, which is used by structural analysis to estimate the structural mass.

The propulsion subsystem calculates expressions for the weight and power requirements of the propulsion, feeding this into the total mass and power modules. SARfunction, which is created by SARsizing, then calculates the power and mass expressions for the SAR system, feeding into total mass and total power. The combined total power is fed into the power subsystem and TMS, which gives expressions of their mass, which is then fed into the total mass subsystem.

The total mass module then gives an optimization expression for the total mass, just in terms of the width, height and nose length optimization variables. This optimization expression is then solved using fmincon and some constraints, including a minimum height, minimum area of SAR for non-ambiguous results, and maximum size of solar panels so that they can be body mounted.

Subsystem code highlights

The propulsion subsystem incorporated a diverse range of engine options, including models like the Busek 100W [10] and Enpulsion Micro R3 [11], to ensure optimal thrust capabilities between 1-3mN. All of the Hall effect thrusters from NASA's state-of-the-art smallsat technology [12] with an available thrust vs. power curve were included. This flexibility enhances adaptability to the best propulsive unit as the drag changes in the optimization. In the Attitude Determination and Control System (ADCS), the output of ADBsat was implemented to account for internal disturbances and asymmetrical mass distribution. The system uses a database of technologically-ready reaction wheels and magnetorquers, optimizing for torque, power, and mass to select the most efficient components.

The Thermal Management Control System (TMCS) is designed to be highly adaptive, calculating heat influx from external sources like the sun and Earth's albedo and internal heat dissipation. It then determines the required size of the radiator and insulation layers, effectively managing the thermal aspects of the satellite. The power subsystem underwent significant refinements to improve its accuracy and efficiency. Using STK simulations, the system now accounts for varying solar incidence angles and provides a more nuanced power requirement calculation for components like the Synthetic

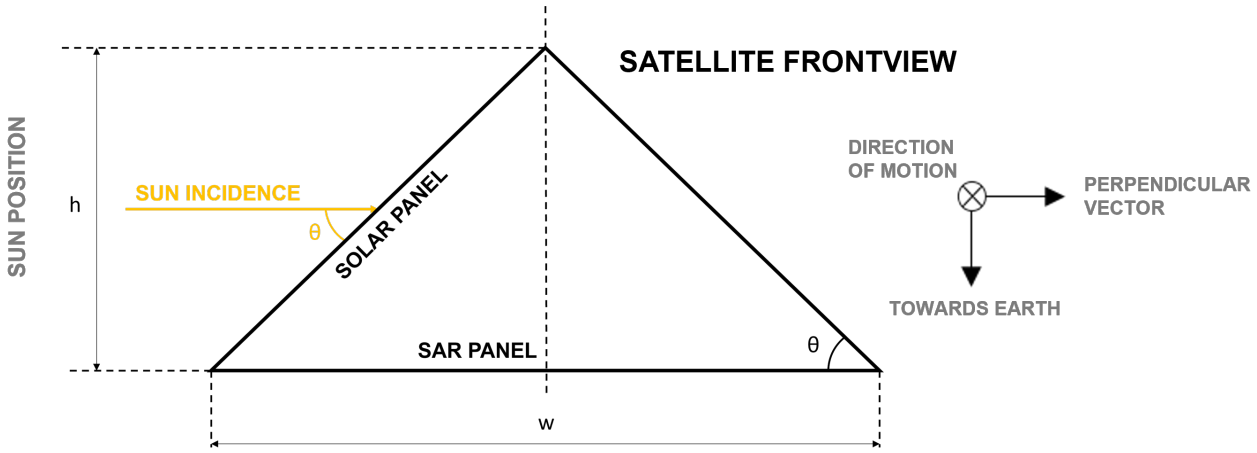

Figure 5. **Diagram showing new incidence of sun after change in orbit.**

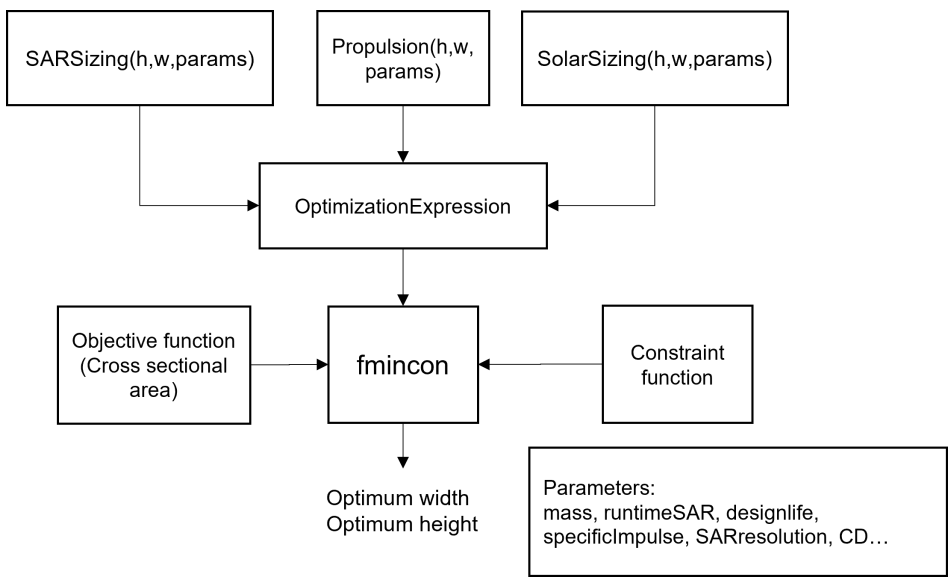

Figure 6. **Initial setup for the optimization algorithm.**

Table 6. **Comparison between initial design in equatorial orbit to first optimization run in sun-synchronous orbit.**

Element	Equatorial Orbit	SSO (Optimized)
Width (m)	0.8265	0.7441
Height (m)	0.4	0.4
Cross-sectional area (m^2)	0.2053	0.1888
Drag force (mN)	1.1	1.0
Total power (W)	240.37	286.36
SAR Area (m^2)	2.48	2.23
Solar Panel Area (m^2)	2.65	1.639
Total mass (kg)	39.33	29.28

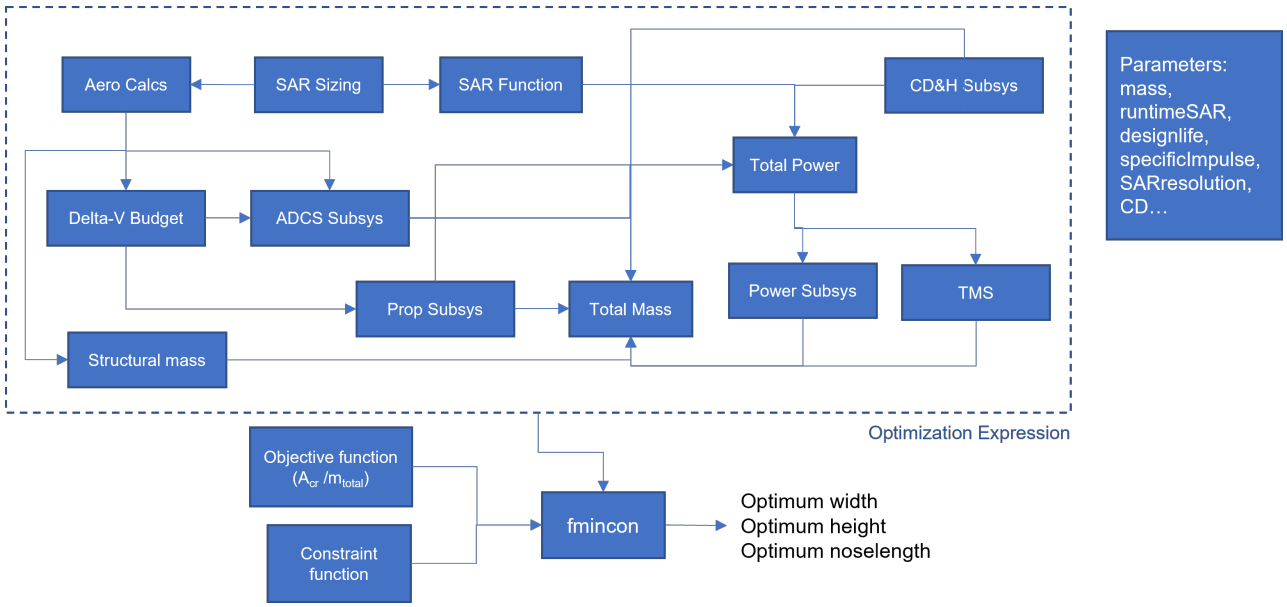

Figure 7. Current optimization algorithm.

Aperture Radar (SAR). The energy storage cycle was analyzed and the most critical battery cycles were identified using a Fourier transform. By increasing the solar panel size slightly, these negative cycles were removed, optimizing the system to minimise mass requirements for batteries and solar panels whilst ensuring accuracy.

Iterations list

Throughout the progression of this study, the amount of subsystems and their level of detail has been increased incrementally. All of the design points were re-run using the final version of the code, to see how the optimization algorithm progressively gets closer to the optimum design. This is shown in Figure 8.

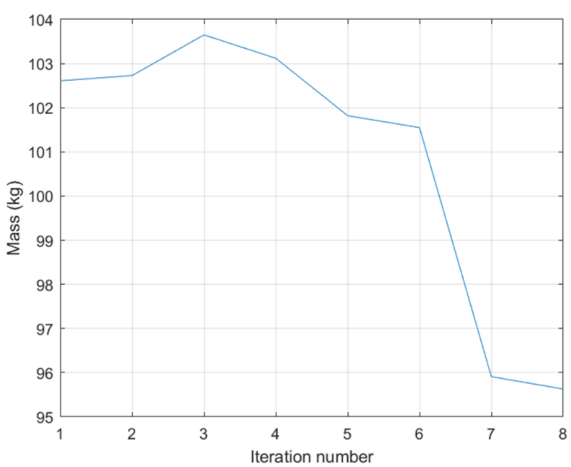

Figure 8. Improvement in mass as optimization algorithm was improved.

As can be observed, the mass generally decreases except for iterations 3 and 4 where the mass increases. This was because as the subsystems were added, those two runs did not include the TMS and had an inaccurate model for the battery sizing. This meant that having a high power was not penalising enough, so that when it was re-run it was further away from the optimal design point. Whilst each iteration progressively decreases the mass, from the first to the last iteration, the mass decrease is only of 7%. Considering that it is unlikely this mass value accurately represents the satellite mass at the late stages, the detail in the optimization algorithm can be considered good and close enough to the optimal point.

Table 7 shows the values at the final iteration. It is interesting to note that, whilst literature suggested that a nose could reduce drag, in our case it turned out that the addition of mass from the nose tended to outweigh its fuel and power savings due to reduced drag. However, the nose could be beneficial in terms of stability in cases where the centre of mass is not sufficiently in front the centre of pressure. Furthermore, the optimization algorithm ensures stability, but only sizes the ADCS for maximum torques that can be expected, assuming the satellite will not rotate around and not considering the extra torque forces that would be needed to overcome for turning. For other mission types, this is something that should be further investigated.

5. Results and Discussion

Key findings

The primary objective of this study was to explore the feasibility of designing a VLEO SAR satellite that is both cost-effective and high-performing. Remarkably, the final optimized design achieved a system mass well below the initial target, coming in at under 100kg. This is particularly noteworthy given that the satellite is capable of achieving resolutions as fine as 0.25m in spotlight mode, with swath widths extending up to 5km. These performance metrics significantly outperform existing commercial SAR satellites in the same weight category, thereby highlighting the untapped potential of VLEO for earth observation missions.

Table 7. **Final iteration key values**

Element	Final iteration
Width (m)	0.6158
Height (m)	0.3565
Nose (mm)	0.4
Cross-sectional area (m^2)	0.1353
C_D	4.323
Drag force (mN)	1.413
Total power (W)	661.8
SAR Area (m^2)	1.847
Solar Panel Area (m^2)	1.682
Total mass (kg)	95.63

One of the key innovations in this study was the development of a comprehensive optimization tool. This tool not only selects the most suitable off-the-shelf components but also provides the optimal shape and calculates the key parameters needed for the detailed design of the phased array. Importantly, this tool is versatile enough to be generalized for other payloads, thereby accelerating the commercial adoption of VLEO orbits.

In terms of aerodynamic challenges, initially considered to be a significant hurdle, our findings suggest that these can be effectively managed. With the advent of electric propulsion technologies, the issue of drag becomes increasingly less significant. Moreover, as air-breathing electric propulsion systems become more readily available, drag could become a non-issue altogether.

Stability remains a concern, particularly in the context of existing reaction wheels and magnetorquers designed for torques outside the atmosphere. However, as VLEO satellites become more common, commercial solutions with higher torques will likely become available. Alternatively, thrust vectoring could be employed to address this issue.

Points of improvement and future work

While most subsystems in the optimization algorithm have been characterized following the literature and can therefore be considered accurate models, this is not true for the structural weight. The current model is empirical and has been fitted by comparing existing satellites to their volume and surface area. The thinking behind this approach was that, even if the structural mass is not accurate, fitting the structural mass in this way would ensure that it scales correctly, maintaining the right optimization gradient. Importantly, this model does not account for any additional structural mass that could result from the use of corrosive-resistant materials. Future research should investigate methods for estimating the structural mass required by a VLEO satellite during the initial stages of development and define a model that can be seamlessly integrated into the current optimization algorithm.

The code should also be generalized to serve as a tool for a broader range of missions. While altering parameters such as altitude, inclination, and other orbit characteristics is already possible, allowing this optimization to accommodate different payloads, such as phased arrays, presents a challenge. For instance, while a SAR has a fixed length to ensure unambiguous results, this is not necessarily the case for other payloads, potentially increasing the number of variables. In the future, the model equations should include more variables so that users can define constraints specific to their payloads. Additionally, subsystems like solar panels, payloads, and thermal management systems have been fixed on one surface of the satellite. This arrangement needs to be reconsidered, or at least more flexible options provided based on the mission type. These aspects require further investigation in the next stage, including verifying whether the optimization algorithm still converges and successfully reaches a minimum point after the added complexity.

Finally, it would be interesting to run the same optimization for the same payload at a range of altitudes, as there could potentially be a higher altitude where the atmospheric density decreases at a faster rate than the weight penalty of the payload. This would provide a more comprehensive view of the advantages of VLEO over LEO and their applicability to different scenarios. Moreover, altitude could also be included as a variable in the optimization process, potentially leading to an even more optimal design point.

Final remarks

In conclusion, this study proves that VLEO SAR satellites are not only feasible but also offer a range of advantages over their LEO counterparts. The optimization tool developed as part of this research could serve as a valuable resource for future studies and commercial applications, potentially revolutionizing the field of earth observation. While challenges related to aerodynamics and stability exist, they are far from insurmountable and are likely to become less significant as technology advances.

This work was sponsored by ST Engineering Satellite Systems, who plan to use this optimization tool for the design and development of upcoming missions. At the time of publication, the company is not planning on making this tool publicly available.

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BIOGRAPHY

Yago Molano Gomez *is currently completing his MEng In Aeronautical Engineering with a Year in Industry at Imperial College London. During the past year, he has been working as a Systems Engineer in the Product and Innovation team of ST Engineering Satellite Systems in Singapore. He has an interest in optimization methods applied to Systems Engineering, but also to other fields such as Trajectory Optimization, which he is investigating for his master's thesis in collaboration with Airbus Space. He is expecting to research these topics further in a PhD.*